

DIYA CHANDA

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Education

The Neotia University <i>B.Tech in Computer Science and Engineering (AIML)</i>	2023 – 2027 CGPA: 9.54
Rajkumari Santanamoyee Girls' High School <i>XII (WBCHSE)</i>	2023 Grade: 82.4%

Experience

AI/ML Research Intern <i>National Institute of Technology (NIT) Durgapur</i>	Jun 2026 – Jul 2026
- Investigated 7 EfficientNet-B0 variants in TensorFlow by replacing standard linear convolutions with quadratic formulations and evaluating RBF, Polynomial, Swish, Leaky ReLU, and ELU activations. - Systematically benchmarked variants against the 78.72% baseline, analyzing vanishing gradients, numerical instability, and activation behavior to identify the most stable quadratic configuration.	
AI/GenAI Intern <i>GenByte AI, Kerala, India</i>	Jun 2026 – Aug 2026 <i>Remote</i>
- Developed a college admission web application using LangChain, Streamlit, and FastAPI to enable users to interact with and retrieve information from admission-related documents. - Built document-processing workflows using PyMuPDF for standard PDF extraction and OCR-based techniques for scanned documents, integrating extracted content into the application's retrieval pipeline.	
Web Development Intern <i>Xeta Labs, Guwahati, India</i>	Jun 2025 – Sep 2025
- Developed and integrated a full LMS system into a production client website across Student, Tutor, and Admin roles. - Debugged and resolved 50+ backend issues pre-launch, covering authentication, REST APIs, and database reliability.	

Projects

CampusSphere — Co-Curricular Governance Platform	Live Demo GitHub
Tech Stack: Next.js, Tailwind CSS, Express.js, PostgreSQL, JWT, bcrypt, Pino. - Contributed to a three-portal Admin, Faculty, and Student platform with 25+ REST APIs supporting activity submissions, approval workflows, credit banking, grievance handling, and institutional reporting. - Implemented authentication, role-based authorization, API workflows, and backend functionality for secure management of student activities and co-curricular records.	
FruitQ-GradeX — AI-Powered Fruit Quality Assessment	Live Demo GitHub
Tech Stack: TensorFlow, Keras, OpenCV, Streamlit, Grad-CAM, NumPy, Pandas. - Developed a multi-head CNN to jointly classify 4 fruit types and quality conditions using 9,146 images, achieving 96.8% and 98.5% test accuracy respectively. - Integrated Grad-CAM for explainable predictions and deployed a Streamlit application supporting real-time image and webcam-based fruit quality assessment.	

Research Publications

- S. Dutta, S. G. Neogi, **D. Chanda**, A. Pramanik, Ö. Girgin and E. L. Öncül, "FruitQ-GradeX: Determining Fruit Quality and Grading with Explainable Deep Learning," *IEEE ICRITO 2025*, pp. 1–7. DOI: 10.1109/ICRITO66076.2025.11241706.
- S. Dutta, S. G. Neogi, S. Chowdhury, V. Chole, A. Pramanik and **D. Chanda**, "Cropsense: Explainable Deep Learning Framework for Accurate Quality Detection in Solanaceous Crops," *IEEE ICRITO 2025*, pp. 1–8. DOI: 10.1109/ICRITO66076.2025.11241535.

Skills

Programming Languages: Python, Java, SQL, C
AI/ML & Deep Learning: PyTorch, TensorFlow, Keras, scikit-learn, OpenCV
Generative AI & NLP: LangChain, RAG, LLMs, Vector Search
Web & Backend: FastAPI, Node.js, Express.js, React.js, REST APIs
Databases & Cloud: PostgreSQL, MongoDB, Supabase, AWS, Redis
Tools & Deployment: Docker, Git, GitHub, GitHub Actions